

第二次月考

题目 1. 已知线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0, \\ \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = 0 \end{cases}$$

的系数矩阵 $A = (a_{ij})$ 满足

(i) $|A| = 0$;

(ii) 某个元素 a_{ij} 的代数余子式 $A_{ij} \neq 0$.

求证：上述线性方程组的解都能写成如下形式：

$$\lambda \begin{bmatrix} A_{i1} \\ \vdots \\ A_{in} \end{bmatrix}.$$

证明. 由 $\sum_{j=1}^n a_{kj}A_{ij} = \begin{cases} 0 & k \neq i \\ |A| & k = i \end{cases}$ 且 $|A| = 0$.

$$\Rightarrow \text{对 } \forall 1 \leq k \leq n, \sum_{j=1}^n a_{kj} A_{ij} = 0.$$

$$\Rightarrow \begin{bmatrix} A_{i1} \\ \vdots \\ A_{in} \end{bmatrix} \text{ 为方程组的解.}$$

$$\text{由 } |A| = 0 \Rightarrow r(A) < n.$$

$$\text{由 } A_{ij} \neq 0 \Rightarrow |A| \text{ 存在 } n-1 \text{ 阶子式不为 } 0 \Rightarrow r(A) \geq n-1.$$

$$\text{从而 } r(A) = n-1.$$

基础解系中包含 $n - r(A) = 1$ 个解向量.

$$\text{又由 } A_{ij} \neq 0 \Rightarrow \begin{bmatrix} A_{i1} \\ \vdots \\ A_{in} \end{bmatrix} \text{ 为非零解向量.}$$

$$\Rightarrow \begin{bmatrix} A_{i1} \\ \vdots \\ A_{in} \end{bmatrix} \text{ 为基础解系.}$$

$$\Rightarrow \text{方程组的解可以写成 } \lambda \begin{bmatrix} A_{i1} \\ \vdots \\ A_{in} \end{bmatrix}.$$

□

题目 2. 已知以下非齐次线性方程组有非零解:

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + \cdots + a_{2n}x_n = b_2, \\ \cdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

求证: 每个解向量中第 k 个分量都等于 0 的充分必要条件是将增广矩阵 $\bar{A}$ 去掉第 k 列后得到的矩阵秩比 $\bar{A}$ 小.

证明.

必要性:

令 $\alpha_i = (a_{1i}, \dots, a_{mi})^\top$, $\beta = (b_1, b_2, \dots, b_m)^\top$.

若 $\bar{A}$ 与去掉第 k 列后得到的矩阵秩相同.

则 $\alpha_1, \dots, \alpha_n, \beta$ 与 $\alpha_1, \dots, \alpha_{k-1}, \alpha_{k+1}, \dots, \alpha_n, \beta$ 等价.

$\Rightarrow \alpha_k$ 可由 $\alpha_1, \dots, \alpha_{k-1}, \alpha_{k+1}, \dots, \alpha_n, \beta$ 线性表出.

设 $\alpha_k = y_1 \alpha_1 + \dots + y_{k-1} \alpha_{k-1} + y_{k+1} \alpha_{k+1} + \dots + y_n \alpha_n + y_{n+1} \beta$.

若 $y_{n+1} \neq 0$.

$$\Rightarrow \beta = -\frac{y_1}{y_{n+1}} \alpha_1 - \dots - \frac{y_{k-1}}{y_{n+1}} \alpha_{k-1} + \frac{1}{y_{n+1}} \alpha_k - \frac{y_{k+1}}{y_{n+1}} \alpha_{k+1} - \dots - \frac{y_n}{y_{n+1}} \alpha_n.$$

则方程组存在一个第 k 个分量不为 0 的解. 矛盾!

若 $y_{n+1} = 0$.

$$\Rightarrow \alpha_k = y_1 \alpha_1 + \dots + y_{k-1} \alpha_{k-1} + y_{k+1} \alpha_{k+1} + \dots + y_n \alpha_n \quad ①$$

设方程组的一个非零解为 $(z_1, \dots, z_{k-1}, 0, z_{k+1}, \dots, z_n)^\top$.

$$\text{则 } z_1 \alpha_1 + \dots + z_{k-1} \alpha_{k-1} + z_{k+1} \alpha_{k+1} + \dots + z_n \alpha_n = \beta \quad ②$$

由 ①, ② 得

$$(z_1 + y_1) \alpha_1 + \dots + (z_{k-1} + y_{k-1}) \alpha_{k-1} - \alpha_k + (z_{k+1} + y_{k+1}) \alpha_{k+1} + \dots + (z_n + y_n) \alpha_n = \beta.$$

方程组有第 k 个分量不为零的非零解. 矛盾!

故 $r(\alpha_1, \dots, \alpha_n, \beta) > r(\alpha_1, \dots, \alpha_{k-1}, \alpha_{k+1}, \dots, \alpha_n, \beta)$.

充分性:

假设 $\exists$ 一个解 $(\omega_1, \dots, \omega_n)^\top$ 满足 $\omega_k \neq 0$.

则 $\omega_1 \alpha_1 + \dots + \omega_{k-1} \alpha_{k-1} + \omega_k \alpha_k + \omega_{k+1} \alpha_{k+1} + \dots + \omega_n \alpha_n = \beta$.

$$\Rightarrow \alpha_k = \frac{1}{\omega_k} (\beta - \omega_1 \alpha_1 - \dots - \omega_{k-1} \alpha_{k-1} - \omega_{k+1} \alpha_{k+1} - \dots - \omega_n \alpha_n).$$

即 α_n 可被 $\alpha_1, \dots, \alpha_{k-1}, \alpha_{k+1}, \dots, \alpha_n, \beta$ 线性表出.

$\Rightarrow \alpha_1, \dots, \alpha_{k-1}, \alpha_k, \alpha_{k+1}, \dots, \alpha_n, \beta$ 与 $\alpha_1, \dots, \alpha_{k-1}, \alpha_{k+1}, \dots, \alpha_n, \beta$ 等价.

$$\Rightarrow r(\alpha_1, \dots, \alpha_{k-1}, \alpha_k, \alpha_{k+1}, \dots, \alpha_n, \beta) = r(\alpha_1, \dots, \alpha_{k-1}, \alpha_{k+1}, \dots, \alpha_n, \beta).$$

矛盾!

故每个解向量中第 k 个分量均为 0. □